

Security Audit

Virovita Token (VVT)

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CAUTION

THIS DOCUMENT IS A SECURITY AUDIT REPORT AND MAY CONTAIN CONFIDENTIAL INFORMATION. THIS INCLUDES IDENTIFIED VULNERABILITIES AND MALICIOUS CODE WHICH COULD BE USED TO COMPROMISE THE PROJECT. THIS DOCUMENT SHOULD ONLY BE FOR INTERNAL USE UNTIL ISSUES ARE RESOLVED. ONCE VULNERABILITIES ARE REMEDIATED, THIS REPORT CAN BE MADE PUBLIC. THE CONTENT OF THIS REPORT IS OWNED BY HASHLOCK PTY LTD FOR USE OF THE CLIENT.

Executive Summary

The ViroVita Token (VVT) team engaged Hashlock Pty Ltd to perform an independent security review of the VVT token's on-chain deployment on the XRP Ledger. The assessment focused on evaluating configuration integrity and alignment with XRPL security best practices and is limited strictly to the token's on-chain parameters at the time of review.

Project Context

ViroVita Token (VVT) is the coordination and incentive layer of the ViroVita infrastructure ecosystem.

VVT operates alongside VVT-Pay, a non-custodial programmable payment orchestration layer built on the XRP Ledger. VVT-Pay enables compliant value flows by orchestrating payment intents and settlement instructions without holding or taking custody of user funds. Initially deployed within healthcare payment and care coordination workflows, VVT is designed to incentivize institutional operators, service providers, and ecosystem participants based on verified participation, performance, and network contribution. As VVT-Pay expands beyond healthcare, VVT applies wherever programmable settlement and coordinated value orchestration are deployed across regulated environments. VVT is structured as a utility asset intended to support ecosystem alignment and operational incentives. It is not designed as a custodial financial product, speculative instrument, or mechanism for guaranteed financial returns. Verified on the XRP Ledger.

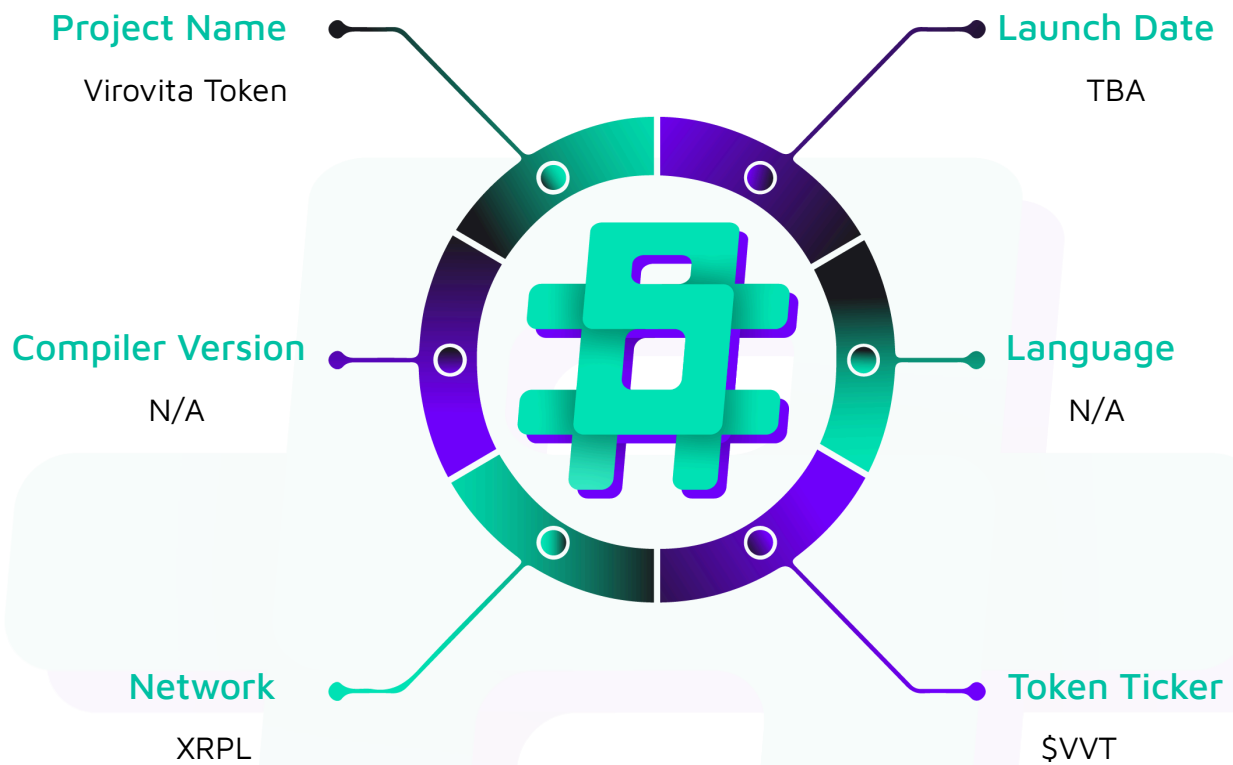
Project Name: Virovita Token

Project Type: Token

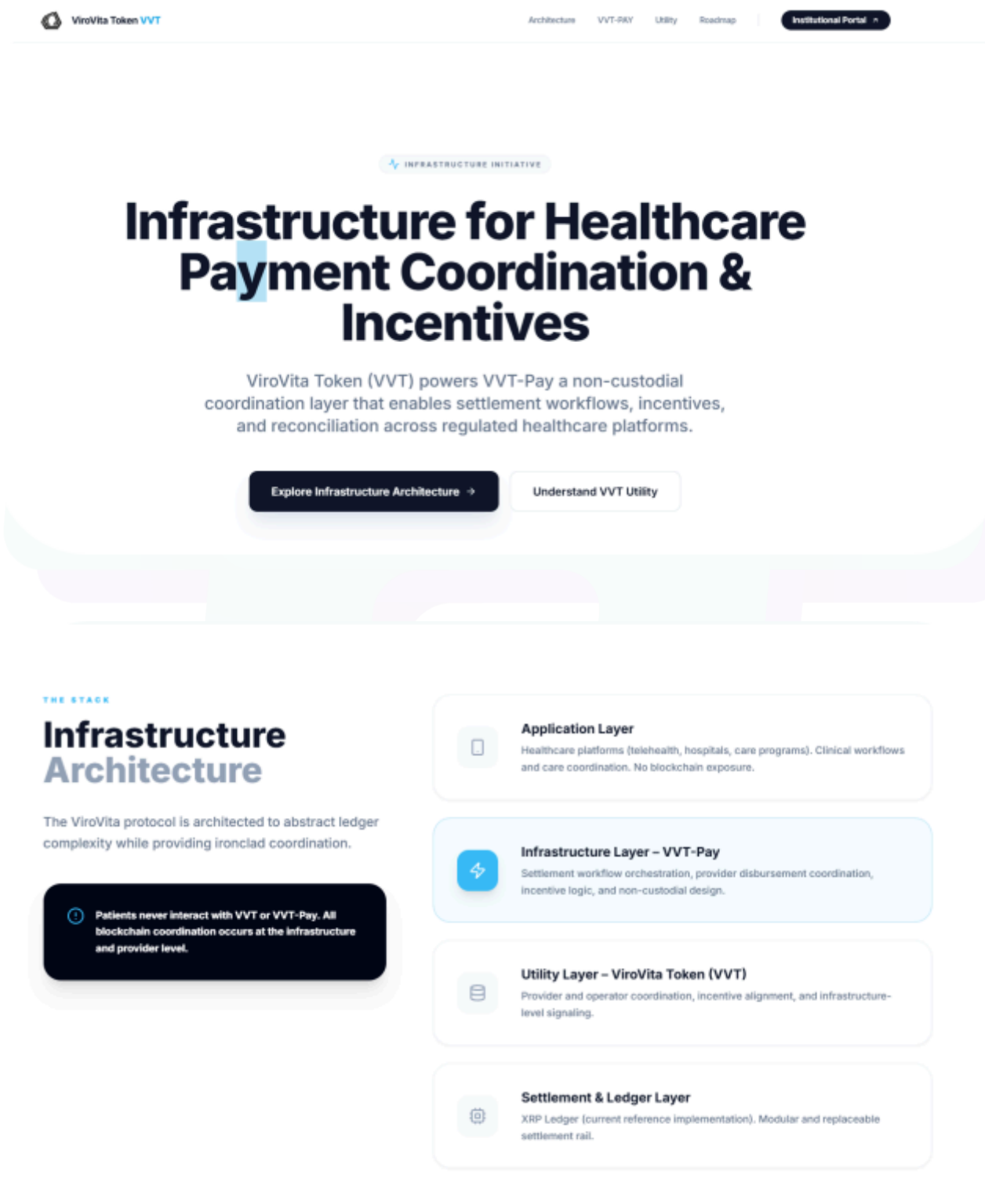
Website: <https://token.virovita.io>

Logo:



Visualised Context:

Project Visuals:



Audit Scope

We at Hashlock audited the token within the Virovita project, the scope of work included a comprehensive review of the token listed below. We tested the token to check for their security and efficiency.

Description	Virovita Token Review
Platform	XRP Ledger
Audit Date	April, 2025
Token Name	Virovita
Token Address	r421CokCfwPxabbdXzCJ77vHkw6eAUWXua
XRPScan URL:	https://xrpscan.com/account/r421CokCfwPxabbdXzCJ77vHkw6eAUWXua

Security Rating

After Hashlock's Audit, we found the token to be **"Secure"**. The token follows simple logic, with correct and detailed ordering.



Not Secure

Vulnerable

Secure

Hashlocked

The 'Hashlocked' rating is reserved for projects that ensure ongoing security via bug bounty programs or on chain monitoring technology.

All issues uncovered during automated and manual analysis were meticulously reviewed and applicable vulnerabilities are presented in the [Audit Findings](#) section. The list of audited assets is presented in the [Audit Scope](#) section and the project's contract functionality is presented in the [Intended Smart Contract Functions](#) section.

All vulnerabilities initially identified have now been resolved and acknowledged.

Hashlock found:

1 Gas Optimisations

1 QA

Caution: *Hashlock's audits do not guarantee a project's success or ethics, and are not liable or responsible for security. Always conduct independent research about any project before interacting.*

Intended Smart Contract Functions

Claimed Behaviour	Actual Behaviour
Virovita Token - A functional Token	Token achieves this functionality.

Code Quality

This audit scope involves the token of the Virovita project, as outlined in the Audit Scope section. All the resources audited follow standard best practices and to help avoid unnecessary complexity that increases the likelihood of exploitation, however, some refactoring was recommended.

Audit Resources

We were given the Virovita Token project token in the form of token address.

The logic is straightforward, and therefore it is easy to quickly comprehend.

Dependencies

As per our observation, this token infrastructure is based on well-known industry standard open source projects.

Severity Definitions

The severity levels assigned to findings represent a comprehensive evaluation of both their potential impact and the likelihood of occurrence within the system. These categorizations are established based on Hashlock's professional standards and expertise, incorporating both industry best practices and our discretion as security auditors. This ensures a tailored assessment that reflects the specific context and risk profile of each finding.

Significance	Description
High	High-severity vulnerabilities can result in loss of funds, asset loss, access denial, and other critical issues that will result in the direct loss of funds and control by the owners and community.
Medium	Medium-level difficulties should be solved before deployment, but won't result in loss of funds.
Low	Low-level vulnerabilities are areas that lack best practices that may cause small complications in the future.
Gas	Gas Optimisations, issues, and inefficiencies.
QA	Quality Assurance (QA) findings are informational and don't impact functionality. Supports clients improve the clarity, maintainability, or overall structure of the code.

Status Definitions

Each identified security finding is assigned a status that reflects its current stage of remediation or acknowledgment. The status provides clarity on the handling of the issue and ensures transparency in the auditing process. The statuses are as follows:

Significance	Description
Resolved	The identified vulnerability has been fully mitigated either through the implementation of the recommended solution proposed by Hashlock or through an alternative client-provided solution that demonstrably addresses the issue.
Acknowledged	The client has formally recognized the vulnerability but has chosen not to address it due to the high cost or complexity of remediation. This status is acceptable for medium and low-severity findings after internal review and agreement. However, all high-severity findings must be resolved without exception.
Unresolved	The finding remains neither remediated nor formally acknowledged by the client, leaving the vulnerability unaddressed.

Audit Findings

Low

[L-01] Virovita - Single-Signer Multisig in Virovita Issuer Account

Description

The Virovita account, an issuer of VVT and SOLO tokens with 16 NFTs and 12 trust lines, uses a single-signer multisig (threshold: 1, signer: `r4TrqURzPsrRtdLyCmjxjskCqUCxVJrmwC`) despite disabling the master key (`IsfDisableMaster: true`). This configuration negates the security benefits of multisig, functioning as a single-key setup.

Impact

A compromised signer key (`r4TrqURzPsrRtdLyCmjxjskCqUCxVJrmwC`) or regular key (`rwvxiopvsnu8433JxatHCSTPg9R7ikNj5A`) could allow unauthorized token issuance, NFT operations, or trust line modifications, impacting users relying on Virovita's assets.

Recommendation

Implement true multisig with multiple signers for distributed control. Make sure the math works something like this:

- Signer Quorum : 5
- Signer 1 weight : 2
- Signer 2 weight : 3
- Signer 3 weight : 4

To reach Quorum of 5, now at least 2 signers must sign to reach the threshold.

Status

Resolved

QA

[Q-01] Sio2AdapterAssetManager - Underfunded Reserve in Virovita Issuer Account

Description

The Virovita account, with `OwnerCount` = 21 (16 NFTs, 12 trust lines, 1 signer list), has a balance of 7.027473 XRP, far below the required reserve of 52 XRP (10 XRP base + $21 * 2$ XRP owner reserve). This discrepancy risks operational restrictions.

Impact

The underfunded reserve may prevent new transactions (e.g., trust line creation, NFT minting), disrupting Virovita's role as an issuer of VVT and SOLO tokens and affecting users and counterparties.

Recommendation

1. Fund the account to at least 52 XRP to meet reserve requirements.
2. Monitor balance to ensure operational continuity.

Status

Resolved

NFT Clarification Statement

Certain non-fungible tokens (NFTs) were previously minted on the XRP Ledger during early-stage technical exploration conducted by the project team. These NFTs constitute separate digital assets and are not technically integrated with, dependent upon, or functionally connected to the ViroVita Token (VVT) configuration assessed within this audit.

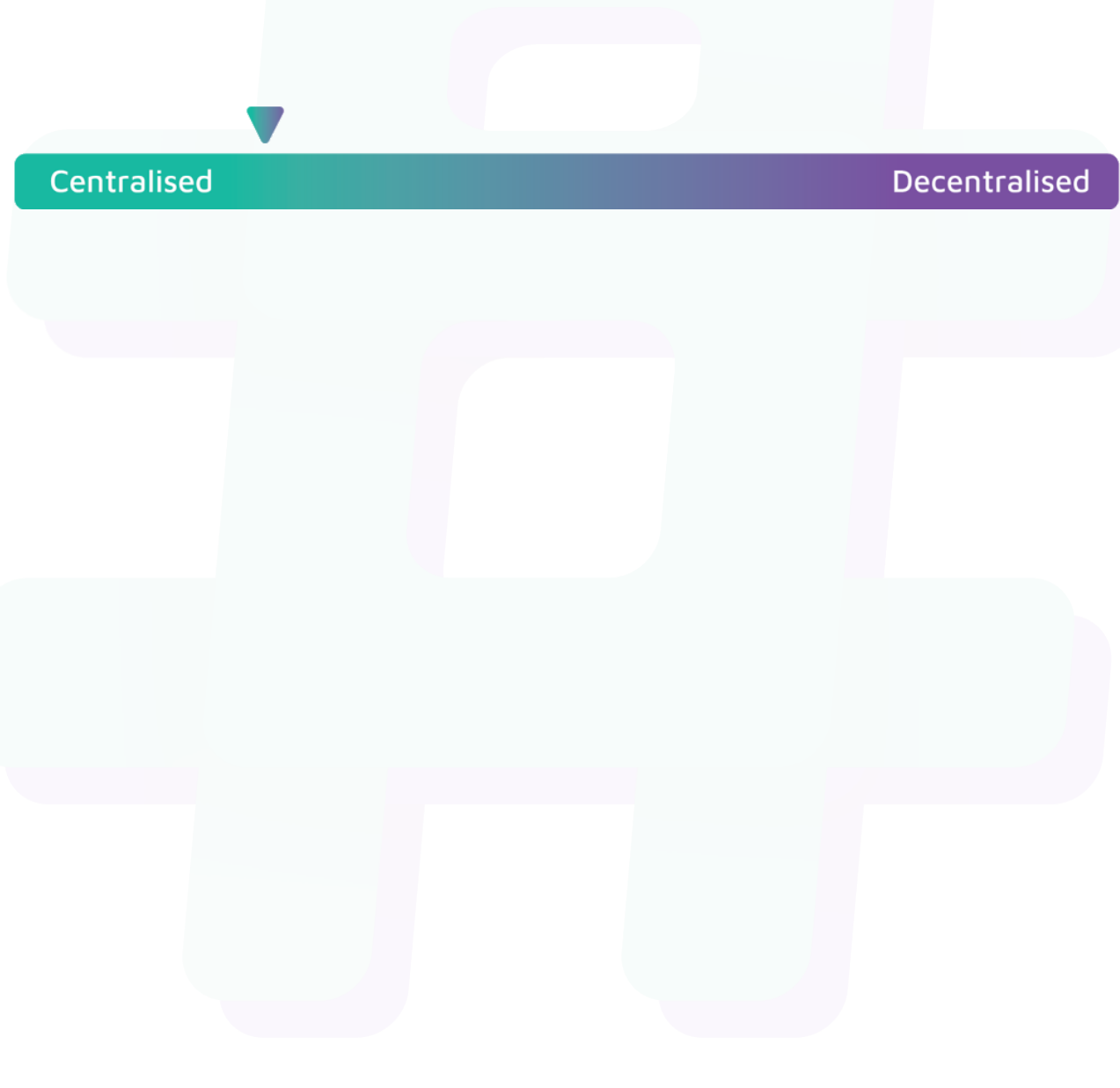
This security assessment applies exclusively to the VVT token deployment and its on-chain configuration at the time of review. It does not evaluate, endorse, or assess any separately issued NFTs, related digital assets, or associated tokenization experiments.

NFT activity does not form part of the current VVT token utility design, tokenomics framework, governance structure, or the VVT-Pay non-custodial programmable infrastructure architecture.

Centralisation

The Virovita Token project values security and efficiency over decentralisation.

Virovita emphasizes a security-focused approach while ensuring operational functionality. By centralizing critical functions, the project enhances reliability and rapid decision-making while maintaining accountability, ensuring the system remains robust and efficient.



Centralised

Decentralised

Conclusion

After Hashlock's analysis, the Virovita Token project seems to have sound and well-tested resources, now that our vulnerability findings have been resolved and acknowledged. Overall, most of the token deployment is correctly ordered and follows industry best practices. To the best of our ability, Hashlock is not able to identify any further vulnerabilities.

Our Methodology

Hashlock strives to maintain a transparent working process and to make our audits a collaborative effort. The objective of our security audits is to improve the quality of systems and upcoming projects we review and to aim for sufficient remediation to help protect users and project leaders. Below is the methodology we use in our security audit process.

Manual Code Review:

In manually analysing all of the code, we seek to find any potential issues with code logic, error handling, protocol and header parsing, cryptographic errors, and random number generators. We also watch for areas where more defensive programming could reduce the risk of future mistakes and speed up future audits. Although our primary focus is on the in-scope code, we examine dependency code and behaviour when it is relevant to a particular line of investigation.

Vulnerability Analysis:

Our methodologies include manual code analysis, user interface interaction, and white box penetration testing. We consider the project's website, specifications, and whitepaper (if available) to attain a high-level understanding of what functionality the smart contract under review contains. We then communicate with the developers and founders to gain insight into their vision for the project. We install and deploy the relevant software, exploring the user interactions and roles. While we do this, we brainstorm threat models and attack surfaces. We read design documentation, review other audit results, search for similar projects, examine source code dependencies, skim open issue tickets, and generally investigate details other than the implementation.

Documenting Results:

We undergo a robust, transparent process for analysing potential security vulnerabilities and seeing them through to successful remediation. When a potential issue is discovered, we immediately create an issue entry for it in this document, even though we have not yet verified the feasibility and impact of the issue. This process is vast because we document our suspicions early even if they are later shown to not represent exploitable vulnerabilities. We generally follow a process of first documenting the suspicion with unresolved questions, and then confirming the issue through code analysis, live experimentation, or automated tests. Code analysis is the most tentative, and we strive to provide test code, log captures, or screenshots demonstrating our confirmation. After this, we analyse the feasibility of an attack in a live system.

Suggested Solutions:

We search for immediate mitigations that live deployments can take and finally, we suggest the requirements for remediation engineering for future releases. The mitigation and remediation recommendations should be scrutinised by the developers and deployment engineers, and successful mitigation and remediation is an ongoing collaborative process after we deliver our report, and before the contract details are made public.

Disclaimers

Hashlock's Disclaimer

Hashlock's team has analysed these smart contracts in accordance with the best industry practices at the date of this report, in relation to: cybersecurity vulnerabilities and issues in the smart contract source code, the details of which are disclosed in this report, (Source Code); the Source Code compilation, deployment, and functionality (performing the intended functions).

Due to the fact that the total number of test cases is unlimited, the audit makes no statements or warranties on the security of the code. It also cannot be considered as a sufficient assessment regarding the utility and safety of the code, bug-free status, or any other statements of the contract. While we have done our best in conducting the analysis and producing this report, it is important to note that you should not rely on this report only. We also suggest conducting a bug bounty program to confirm the high level of security of this smart contract.

Hashlock is not responsible for the safety of any funds and is not in any way liable for the security of the project.

Technical Disclaimer

Smart contracts are deployed and executed on a blockchain platform. The platform, its programming language, and other software related to the smart contract can have their own vulnerabilities that can lead to attacks. Thus, the audit can't guarantee the explicit security of the audited smart contracts.

About Hashlock

Hashlock is an Australian-based company aiming to help facilitate the successful widespread adoption of distributed ledger technology. Our key services all have a focus on security, as well as projects that focus on streamlined adoption in the business sector.

Hashlock is excited to continue to grow its partnerships with developers and other web3-oriented companies to collaborate on secure innovation, helping businesses and decentralised entities alike.

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#hashlock.